

Mark Jennings

Robotics Engineer · markjennings97@gmail.com · makr.org

Summary

Robotics engineer with 7+ years of experience developing integrated hardware from concept to customer. Hands-on background in mechanical design, electric actuation, software development, system validation, and field deployment.

Experience

Apptronik

July 2026 - Present

Senior Hardware Integration Engineer

- Own bring-up and validation of next-generation humanoid hardware, integrating mechanical, electrical, and software interfaces and performing sub-system endurance testing.
- De-risk long-horizon autonomous behaviors with representative hardware prior to full system bring-up.
- Establish in-house calibration capabilities to improve repeatability and characterization of custom perception stack.

Elevate Robotics (Apptronik spin-off)

September 2024 - June 2026

Robotics Engineer

- Designed a lightweight, high-payload robotic manipulator with an industry-leading >1:1 payload-to-weight ratio from concept to an upcoming product, driving mechanical architecture, structural analysis, actuation, packaging, manufacturability, safety compliance, and cost optimization.
- Led technical analysis for a NASA SBIR effort and secured \$1.275M in Phase II funding by translating robotic system capabilities and development plans into a compelling technical proposal.
- Deployed prototype robotic hardware to construction sites and trade shows, supporting field integration, debugging, and feedback into subsequent design iterations.
- Initiated 4 patents for concepts ranging from industrial design to a novel fail-safe braking system.

Los Alamos National Laboratory

October 2021 - August 2024

Research & Development Engineer

- Overhauled a nuclear manufacturing glovebox with the first autonomous robotic arm used in U.S. plutonium part production, integrating the manipulator with a custom end-effector, alignment sensors, pneumatic rotary table, and operator workflows in a highly regulated and safety critical environment.
- Created control software, operating procedures, and custom tooling for multiple nuclear manufacturing processes, taking systems from conceptual ideas to daily operational use.
- Coordinated multi-year intern program to deploy heterogeneous robot fleet for mock radiation survey.

Nuclear and Applied Robotics Group

August 2019 - August 2021

Graduate Research Assistant

- Researched physical human-robot interaction, augmenting human dexterity/reasoning with robotic precision/strength.
- Developed custom ROS packages in C++ for gravity-compensated manipulator (precursor to Elevate Arm), custom mobile manipulator, and camera calibration.

Skills

- **Mechanical Design:** CAD (Onshape, SolidWorks, NX, Creo), GD&T, DFMA, FEA, linkage/mechanism optimization, industrial design, human factors
- **Software & Algorithms:** C++, Python, Linux, Git, ROS/ROS2, physics simulation, manipulator motion planning, mobile robot localization/navigation/mapping, camera calibration
- **Electric Actuation:** Rotor/stator selection, motor drivers, linear/rotary transmission, reduction (planetary, strain wave, prismatic), brakes, bearings, encoders, controls
- **Fabrication:** CNC/manual machining, additive manufacturing, casting, injection molding, extruding, sheet metal bending, welding, rapid prototyping, general shop and hand tool operation
- **Testing & Validation:** Testbed design, instrumentation, environmental testing, endurance testing, failure analysis

Education

The University of Texas at Austin

Master of Science in Mechanical Engineering (3.96)

August 2021

Bachelor of Science in Mechanical Engineering (3.84)

May 2019